

Wireless Vibrant Virtual Walker: Wireless Foot Vibration Device for Virtual Walking Experience

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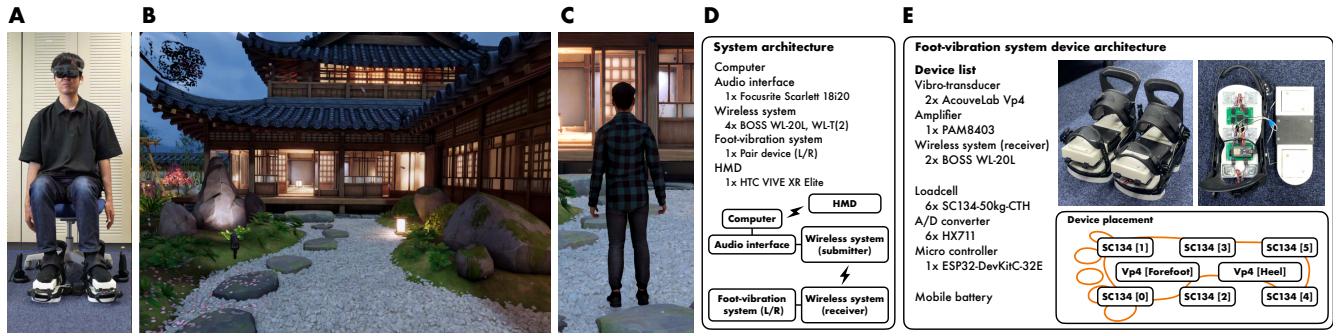


Figure 1: System setup: A. Seated observer with HMD and foot-vibration system. B. Observer's first-person view. C. Observer's self-avatar in the virtual environment. D. System architecture including computer, audio interface, wireless system, and HMD. E. Foot-vibration system with vibro-transducers, load cells, and microcontroller.

Abstract

This research proposes a wireless foot vibration device designed to enhance virtual walking experiences in virtual environments. The device, constructed from snowboard bindings and 3D-printed parts, features vibrators and pressure sensors to provide tactile feedback and control walking direction. Users can navigate virtual environments, created in Unity, by shifting their weight without moving their legs. This system provides a realistic walking sensation through synchronized foot vibrations corresponding to various ground textures. The compact design offers high immersion and freedom, suitable for use in standing, seated, or lying down postures, overcoming physical constraints of traditional systems.

CCS Concepts

• **Human-centered computing** → **Virtual reality; Haptic devices.**

Keywords

Interaction technologies, Multi-/cross-modality, Locomotion interface, Walking, Tactile sensation, Foot vibration, Pressure sensor

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1 Introduction

Walking is the most familiar and widely accepted form of human locomotion. Previous studies have proposed various systems to present virtual walking experiences in virtual environments [Nilsen et al. 2018]. For example, the use of treadmills or low-friction shoes has been proposed. These provide a high-quality walking experience, but they can be expensive, require specific physical spaces, and impose certain physical conditions or constraints on the user.

From this research background, we have developed a new system utilizing a head-mounted display (HMD) and devices worn on the feet to achieve a more flexible and controllable walking experience. Users can move in any direction without moving their legs, simply by shifting their weight on their feet. This allows users to choose stable postures, such as sitting, avoiding spatial and physical constraints. Additionally, their hands remain free for other tasks and interactions, as they are not occupied by the locomotion process.

Foot vibrations are effective in enhancing the sensation of walking [Terziman et al. 2012; Turchet et al. 2012]. By combining foot vibrations with the observation of a self-avatar [Matsuda et al. 2021] and the presentation of the ground texture through vibrations [Nakamura et al. 2021], the virtual walking experience can be further enhanced. Our system integrates these elements with walking control by shifting weight on the feet, allowing users to navigate within a virtual environment with minimal body movement. Furthermore, the device is wireless, enhancing usability and flexibility. To the best of our knowledge, no previous work has fully combined these features into a single system.

The primary contributions of this research are as follows: Firstly, we have designed the system to reduce physical constraints, allowing it to be used in both standing and seated positions. Secondly, we provide a more realistic walking experience by presenting synchronized foot vibrations corresponding to the avatar's walking motion and ground texture. Thirdly, we enable intuitive directional control through weight shifts detected by pressure sensors, offering users a natural navigation environment.

2 System Overview

We designed a wireless foot vibration device. This device is based on snowboard bindings and constructed with 3D-printed parts, making it a DIY-friendly design. It is intended to be affordable, customizable, and constructed without the need for specialized parts. The construction methods and data for the device will be made publicly available.

The device comprises vibrators (AcouveLab Vp4) positioned at the forefoot and heel, an amplifier to drive the vibrators, pressure sensors, and a microcontroller (ESP32-DevKitC-32E) that processes and transmits sensor data. The pressure sensor array consists of six load cells (SC134-50kg-CTH) placed two each at the forefoot, midfoot, and heel. Each load cell outputs data to the ESP32 microcontroller via an A/D converter (HX711) with a resolution of 24 bits and a sampling rate of 10 Hz, capable of measuring up to 50 kg. These were driven by a battery attached to the device (Figure 1-E).

The ESP32 microcontroller connects to the computer via Bluetooth and behaves as an HID-compliant gamepad (Figure 1-D). The device maps the sensor data to XY-axis output. The X-axis is derived by adding the values of the three sensors on the left and right sides and dividing them. The Y-axis is derived by dividing the sum of the forefoot sensor values by the sum of the heel sensor values. The XY-axis output is limited to a range of -5 kg (lower limit) to 5 kg (upper limit). A pair of devices for the left and right feet is integrated within Unity to control the avatar's walking and rotation. Unity C-sharp scripts sum the X and Y data from the pair, treating them as a single virtual device. The X-axis controls the avatar's rotation each frame, with the avatar's Y-axis rotation = input value $\times$ rotation ratio $\times$ (time/frame). The Y-axis controls the avatar's walking animation each frame, with the avatar walking animation blend (-1 to +1; backward, idle, forward) = input value $\times$ speed ratio. Walking animations include Walk Backwards, Idle, and Walk2 from FinalIK. The walking speed is -1.556 [m/s] at -1, 1.556 [m/s] at 1, and stationary at 0. Walking speed changes based on foot pressure, and rotation can be combined. Since walking animation and rotation are independent, either can be controlled separately.

By wearing the HMD and foot vibration devices, users can experience virtual walking (Figure 1-A). The virtual walking environment is developed in Unity and can utilize CG or 360-degree video. Users observe walking from a first-person perspective of the avatar (Figure 1-B, C), and vibrations are presented by the vibrators synchronized with the avatar's foot movements. The type of vibration changes according to the virtual grounds, representing the textures of wood, tatami, stone, gravel, and grass surfaces. There is no restriction on the posture for experiencing virtual walking, allowing users to be standing, sitting, or lying down.

3 Demonstration

The virtual environment can be selected from either a CG world created in Unity or 360-degree videos. Users can experience virtual walking in either a standing or seated posture. The CG virtual environment includes various ground surfaces such as soil, gravel, grass, and rocks, with corresponding vibrations presented to the soles of the feet as users walk, allowing them to feel changes in the walking environment. In the 360-degree video virtual environment, the ground textures are presented, and a self-avatar is superimposed onto the video, enhancing immersion.

Our developed foot vibration device allows users to have a realistic virtual walking experience, making movement within the virtual environment feel more natural. By shifting their weight on their feet without moving their legs, users can freely control their self-avatar to walk, controlling both direction and speed. This compact system provides a high degree of immersion and freedom in walking experiences, overcoming the physical constraints that previous systems could not achieve.

4 Conclusion

We have developed and demonstrated a wireless foot vibration device to realize virtual walking experiences. This device provides control and feedback for virtual walking through vibrators and pressure sensors. The vibrators at the forefoot and heel present vibrations synchronized with walking and the texture of the ground being walked on. The six pressure sensors capture changes in foot pressure to control walking direction. Users can experience this without being limited by physical constraints, whether in standing, seated, or lying down postures.

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